

U.G.C. NET Exam

COMPUTER

(Solved with Explanation)

1. If the time is now 4 O'clock, what will be the time after 101 hours from now?

- (a) 9 O'clock (b) 8 O'clock
(c) 5 O'clock (d) 4 O'clock

Ans : (a) Current time = 4 O'clock

Time to come back at same position = 12 hour

So clock will be completed cycles in 101 hour and remaining hour is $\rightarrow 101 \bmod 12$

$$= 5 \text{ hour}$$

So, time after 101 hours = $4+5 = 9$ O'clock

2. Let $m = (313)_4$ and $n = (322)_4$. Find three base 4 expansion of $m + n$.

- (a) $(635)_4$ (b) $(32312)_4$
(c) $(21323)_4$ (d) $(1301)_4$

Ans : (d) This question can also be solved by first converting both the number in decimal system then performing addition and then finally converting addition result into base 4.

$$(313) = 3 \times 16 + 4 \times 1 + 3 = 55;$$

$$(322) = 3 \times 16 + 4 \times 2 + 2 = 58;$$

$$\text{addition is } 55 + 58 = 113$$

now 113 in base 4 can be represented as 1301 to the base 4.

3. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ Find the

Boolean product $A \odot B$ of the two matrices.

(a) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

Ans : (a) $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

we have to perform logical AND logical or in place of multiplication and addition for ex. consider following matrix Boolean product.

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R SCIENCE
 (Explanation Paper-II)

Now,

$$A \odot B = \begin{bmatrix} 1 \vee 0 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 0 \vee 0 \\ 0 \vee 0 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 0 \vee 0 \\ 1 \vee 0 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 1 \vee 0 & 0 \vee 0 \vee 0 \\ 0 \vee 0 \vee 1 & 0 \vee 0 \vee 0 & 0 \vee 0 \vee 1 & 0 \vee 0 \vee 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

So, option (a) is correct.

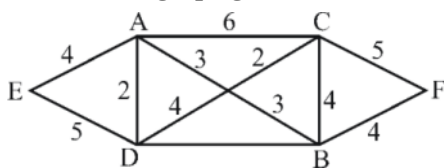
4. How many distinguishable permutations of the letters in the word BANANA are there?

- (a) 720 (b) 120
 (c) 60 (d) 360

Ans : (c) Number of permutation of n objects with n_1 identical object of type 1, n_2 identical object of type 2 & n_k identical objects of type k is $\frac{n!}{n_1!n_2! \dots n_k!}$

Permutation of latter BANANA are $\frac{6!}{3!2!1!} = 60$

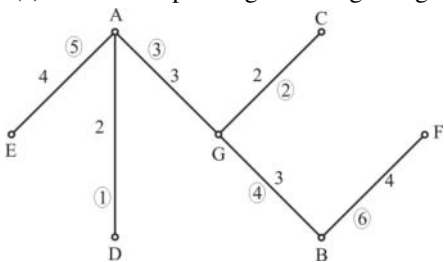
5. Consider the graph given below :



Use Kruskal's algorithm to find a minimal spanning tree for the graph. The list of edges of the tree in the order in which they are chosen is?

- (a) AD, AE, AG, GC, GB, BF
 (b) GC, GB, BF, GA, AD, AE
 (c) GC, AD, GB, GA, BF, AE
 (d) AD, AG, GC, AE, GB, BF

Ans : (c) Minimum spanning tree for given graph:



Kruskal's algorithm start with picking edge with smallest edge weight i.e. "AD = 2" or "GC = 2".

After that “AD = 2”.

Since for 1st choice we choose “AD = 2”.

So, order will GC, AD, GB, GA, BF, EA.

Note: There are more than 1 order possible since edge weights are repeated.

6. The Boolean function with the Karnaugh map

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	0

- (a) $(A + C) \cdot D + B$ (b) $(A + B) \cdot C + D$
(c) $(A + D) \cdot C + B$ (d) $(A + C) \cdot B + D$

Ans : (a)

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	0

So, SoP from = $AD + CD + B$

$$= (A + C) \cdot D + B$$

7. The Octal equivalent of the binary number 1011101011 is :

- (a) 7353 (b) 1353
(c) 5651 (d) 5657

Ans : (b) To solve this (1011101011) question first we make pair of three digit from right side

i.e. 011 011 101 011

011 = 3, 101 = 5, 011 = 3, 011 = 1

So, (001 011 101 011) = (1353) in octal

8. Let P and Q be two propositions, $\neg(P \leftrightarrow Q)$ is equivalent to :

- (a) $P \leftrightarrow \neg Q$ (b) $\neg P \leftrightarrow Q$
(c) $\neg P \leftrightarrow \neg Q$ (d) $Q \rightarrow P$

Ans : (a, b)

P	Q	$\sim P$	$\sim Q$	$Q \rightarrow P$	$\sim(P \leftrightarrow \sim Q)$	$\sim P \leftrightarrow Q$	$\sim P \leftrightarrow \sim Q$	$\sim P \leftrightarrow \sim \sim Q$
0	0	1	1	1	0	0	0	1
0	1	1	0	0	1	1	1	0
1	0	0	1	1	1	1	1	0
1	1	0	0	1	0	0	0	1

Given statement: $\neg(P \leftrightarrow Q)$

We know $A \leftrightarrow B = A'B' + AB$

So, $\neg(P'Q' + PQ)$

$$(P'Q + PQ') \cong P \leftrightarrow \neg Q \cong \neg P \leftrightarrow Q$$

9. Negation of the proposition $\exists x H(x)$ is :

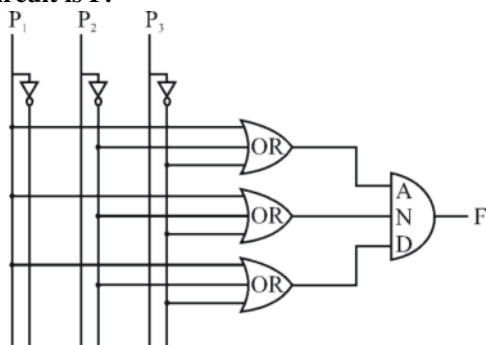
- (a) $\exists x \neg H(x)$ (b) $\forall x \neg H(x)$
(c) $\forall x \neg H(x)$ (d) $\neg x H(x)$

Ans : (b) Proposition $\Rightarrow \exists x H(x)$

Negation $\Rightarrow \neg[\exists x H(x)]$

$\Rightarrow \forall x \neg H(x)$

10. The output of the following combinational circuit is F.



The value of F is :

- (a) $P_1 + P_2 P_3$ (b) $P_1 + P_2 P_3$
(c) $P_1 + P_2 P_3$ (d) $P_1 + P_2 P_3$

Ans : (b) We know that $(A + B)(A + C) = A + BC$ on the basis of given question or gates will produce output $(P_1 + P'_2 + P'_3)$, $(P_1 + P'_2 + P_3)$, $(P_1 + P_2 + P_3)$ and gate will take all three input produced by or gate generate $(P_1 + P'_2 P'_3)$ as output.

11. 'ptrdata' is a pointer to a data type. The expression *ptrdata++ is evaluated as (in C++) :

- (a) $*(ptrdata++)$
(b) $(*ptrdata)++$
(c) $*(ptrdata)++$
(d) Depends on compiler

Ans : (a) Post increment (++) has higher precedence than dereference (*). Hence it increments the address that the pointer holds, then dereferences it.

12. The associativity of which of the following operators operators is left to right, in C++?

- (a) Unary Operator
(b) Logical not
(c) Array element access
(d) Address of

Ans: (a) **Unary operators:** Logical not, one's complement, unary minus, increment, decrement, address of indirection, cast and size of operator are left to left associative while operator function call, array subscript, dot and arrow are left to right associative.

13. A member function can always access the data in , (in C++)

- (a) the class of which it is member
(b) the object of which it is a member
(c) the public part of its class
(d) the private part of its class

Ans : (a) Member function can always access the data in the class of which is member in C++.

Code:-

```
class x {  
Private String name;  
Private int age;  
x (string name, int age)  
{  
this . name = name;  
this . age = age;  
}  
int get Age ()  
{  
return age;  
}  
}
```

14. Which of the following is not correct for virtual function in C++?

- (a) Must be declared in public section of class
- (b) Virtual function can be static
- (c) Virtual function should be accessed using pointers
- (d) Virtual function is defined in base class

Ans : (b) Virtual function cannot be global or static because, a virtual function is a member function of a base class and relies on a specific object to determine which implementation of function is called. You can declare a virtual function to be a friend of another class.

15. Which of the following is not correct (in C++)?

- (a) Class templates and function templates are instantiated in the same way
- (b) Class templates differ from function templates in the way they are initiated
- (c) Class template is initiated by defining an object using the template argument
- (d) Class templates are generally used for storage classes

Ans : (a) **Template** are a feature of the C++ programming language that allow functions and classes to operate with generic types, which allow a function or class to work on many different data type without being rewritten for every one separately.

1. Function template: Behave like function except that the template can have argument of many different types.

- (i) Function template does not occupy space in memory.
- (ii) The function template does not save memory.
- (iii) The function template can be instantiated with any copy constructible type for which the expression is valid.

2. Class template: Provides a specification for generating classes based on parameters.

- (i) They are used commonly to implement containers.
- (ii) A class template is instantiated by passing a given set of types to it as template arguments.

16. Which of the following is/are true with reference to 'view' in DBMS?

- (A) A 'view' is a special stored procedure executed when certain event occurs
 (B) A 'view' is a virtual table, which occurs after executing a pre-compiled query.

Code :

- (a) Only (A) is true
 (b) Only (B) is true
 (c) Both (A) and (B) are true
 (d) Neither (A) nor (B) are true

Ans : (b) View is a virtual table, through which a selective portion of the data from one or more table can be seen. A view does not contain data of its own.

17. In SQL is an Aggregate function.

- (a) SELECT (b) CREATE
 (c) AVG (d) MODIFY

Ans : (c) AVG function returns the average of the values in a group. It ignores null values. It is an Aggregate function.

Syntax : ([All Distinct] expression)

18. Match the following with respect to RDBMS :

A.	Entity integrity	i.	Enforces some specific business rule that to not fall into entity or domain
B.	Domain integrity	ii.	Rows can't be deleted which are used by other records
C.	Referential integrity	iii.	Enforces valid entries for a column
D.	User defined integrity	iv.	No duplicate rows in a table

Code :

- | | A | B | C | D |
|-----|----------|----------|----------|----------|
| (a) | iii | iv | i | ii |
| (b) | iv | iii | ii | i |
| (c) | iv | ii | iii | i |
| (d) | ii | iii | iv | i |

Ans : (b) : Entity integrity is used to provide primary key mechanism hence there is no duplicate rows in table
 : Domain Integrity specifies that all columns in a relational database must be declared upon a defined domain

: Referential integrity means we cannot delete records in master table before deleting the records from child table.

19. In RDBMS, different classes or relations are created using technique to prevent modification anomalies.

- (a) Functional dependencies
 (b) Data integrity
 (c) Referential integrity
 (d) Normal forms

Ans: (d) Normal forms are defined as functional dependency, data integrity, referential integrity, normalization.

20.SQL command changes one or more fields in a record.

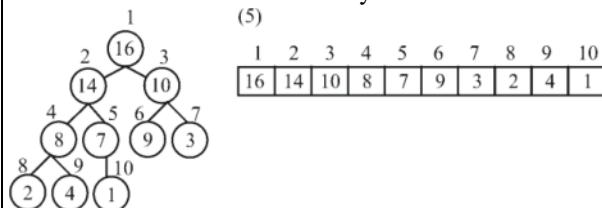
- (a) LOOK-UP (b) INSERT
(c) MODIFY (d) CHANGE

Ans : (c) In DBMS, modify, a SQL command that changes one or more fields in a record.

21. Consider an array representation an n element binary heap where the elements are stored from index 1 to index n of the array. For the element stored at index i of the array ($i \leq n$), the index of the parent is :

- (a) floor $((i + 1)/2)$ (b) ceiling $((i + 1)/2)$
(c) floor $(i/2)$ (d) ceiling $(i/2)$

Ans : (c) Consider a binary heap of n elements in which elements are stored in array from index 1 to n.

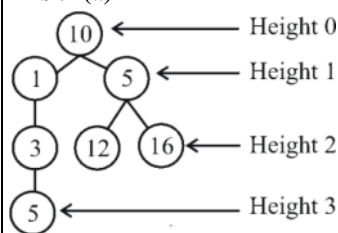


Parent of any element at index i is calculated as $\left\lfloor \frac{i}{2} \right\rfloor$

22. The following numbers are inserted into an empty binary search tree in the given order : 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree?

- (a) 3 (b) 4
(c) 5 (d) 6

Ans : (a)

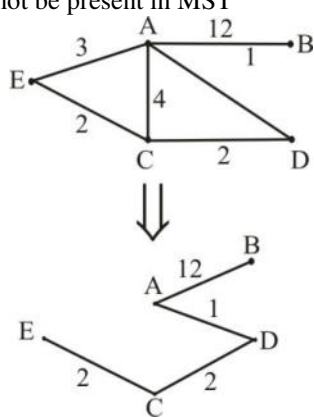


23. Let G be an undirected connected graph with distinct edge weight. Let $E_{\max}$ be the edge with maximum weight and $E_{\min}$ the edge with minimum weight. Which of the following statements is false?

- (a) Every minimum spanning tree of G must contain $E_{\min}$
(b) If $E_{\max}$ is in minimum spanning tree, then its removal must disconnect G.
(c) No minimum spanning tree contains $E_{\max}$
(d) G has a unique minimum spanning tree

Ans : (c) There may exist some spanning tree which consist of $E_{\max}$ in it & its removal must disconnect of the graph G.

It cannot always be the case that the edge with $e_{\max}$ should not be present in MST



$E_{\max}$ is present in MST.

24. A list of n strings, each of length n , is sorted into lexicographic order using merge - sort algorithm. The worst case running time of this computation is :
- (a) $O(n \log n)$ (b) $O(n^2 \log n)$
 (c) $O(n^2 + \log n)$ (d) $O(n^3)$

Ans : (b) Since list contain ' n ' string, each of length ' n '. Take 1st character of each string and sort it and place sorted strings into output list, doing this for n times will sort the all the strings into one list, so time complexity = $n \cdot (n \log n) = O(n^2 \log n)$
Note: First sort each string individually in list will take $O(n \log n)$ time more.

25. Post order traversal of a given binary search tree T produces following sequence of keys :
 Which one of the following sequences of keys can be the result of an in-order traversal of the tree T ?
- (a) 3, 4, 5, 7, 9, 14, 20, 18, 17, 16, 15
 (b) 20, 18, 17, 16, 15, 14, 3, 4, 5, 7, 9
 (c) 20, 18, 17, 16, 15, 14, 9, 7, 5, 4, 3
 (d) 3, 4, 5, 7, 9, 14, 15, 16, 17, 18, 20

Ans : (d) In order traversal always gives sorted list in BST. 3, 4, 5, 7, 9, 14, 15, 16, 17, 18, 20 .
 • The in-order traversal of the Binary Search Tree (BST) always given elements in ascending or increasing order.
 So, In order traversal of the sequence of keys given in post- order traversal in sorted ascending order of that keys.

26. Which of the following devices takes data sent from one network device and forwards it to the destination node based on MAC address?
- (a) Hub (b) Modem
 (c) Switch (d) Gateway

Ans : (c) Switch takes data sent from one network device & forwards it to the destination node based on MAC address.

A switch is a multi port bridge with a buffer and a design that can boost its efficiency and performance. Switch is data link layer device. It can perform error checking before forwarding data.

27. do not take their decisions on measurements or estimates of the current traffic and topology.

- (a) Static algorithm
- (b) Adaptive algorithm
- (c) Non-adaptive algorithm
- (d) Recursive algorithm

Ans : (c) Non adaptive algorithm Implements static routing non adaptive algorithm do not base their routing decision on measurements and estimates of the current traffic & topology. Instead the route to be taken in going from one node to the other is computed in advance.

28. The number of bits used for addressing in Gigabit Ethernet is

- (a) 32 bits
- (b) 48 bits
- (c) 64 bits
- (d) 128 bits

Ans : (b)

Ethernet	Frame
62 bits	Preamble used for bit synchronization
2 bits	Start or frame delimiter
48 bits	Destination Ethernet address
48 bits	Source ethernet address
16 bits	Length
46 - 1500 bytes	Data
32 bits	Frame check sequence

The Ethernet frame is the packet that delivers data between stations. Each frame is made up of bits. Which are organized into different fields. The four main fields are the destination Ethernet address field, source. Ethernet address field, data field & frame check sequence field. The address fields are 48 bit long.

29. Which of the following layer of OSI Reference model is also called end-to-end layer?

- (a) Network layer
- (b) Datalink layer
- (c) Session layer
- (d) Transport layer

Ans : (d) Transport layer is the layer in the OSI model responsible for end to end communication over a network. It provides logical communication between application a layered architecture of protocols and other network component.

30. The IP address is used by hosts when they are being booted.

- (a) 0.0.0.0
- (b) 1.0.0.0
- (c) 1.1.1.1
- (d) 255.255.255.255

Ans : (a) 0.0.0.0 is also called DHCP address used when host was booted first time.
In IPV4 0.0.0.0 is used for the default route.

31. Consider the following program fragment in assembly language :

```

mov ax, 0h
mov cx, 0A, h
doloop :
    dec ax
    loop doloop

```

What is the value of ax and cx registers after the completion of the doloop?

- (a) ax = FFF5 h and cx = 0 h
- (b) ax = FFF6 h and cx = 0 h
- (c) ax = FFF7 h and cx = 0A h
- (d) ax = FFF5 and cx = 0A h

Ans : (b)
dec ax =

1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0
-1
1 1 1 1 1 1 1 1

now, loop do loop

cx = 0 0 0 0 1 0 1 0
ax = 1 1 1 1 1 1 1 1
-1
-1

(1)

0 0 0 0 1 0 1 0
1 1 1 1 1 1 1 0

cx = 0 0 0 0 1 0 0 1
ax = 1 1 1 1 1 1 1 0
-1
-1

(2)

0 0 0 0 1 0 0 0
1 1 1 1 1 1 0 1

now, do as same for 10 time & in the last step is.

cx = 0 0 0 0 0 0 0 1
-1
0 0 0 0 0 0 0 0

cx = 0h so the loop stops here
ax = 11110110
i.e. F6n
Which could be written as FFF6 h

32. Consider the following assembly program fragment :

```

stc
mov al, 110101110b
mov cl, 2
rcl al, 3
rol al, 4
shr al, cl
mul cl

```

The contents of the destination register ax (in hexadecimal) and the status of Carry Flag (CF) after the execution of above instructions, are :

- (a) ax = 003CH; CF = 0
- (b) ax = 001EH; CF = 0
- (c) ax = 007BH; CF = 1
- (d) ax = 00B7H; CF = 1

Ans : (a) If you multiply any binary number by 2, Simply add a 0 bit to its L.S.B. (Right End).

Here, $d = 2$, so we are multiplying the binary number isal by 2.

Hence, $al = 000111100$

Now, it becomes $0000000000111100 = 003CH$

$CF = 0$

$CF = 0$, if high byte/word/double word = 0

$CF = 1$, if high byte/word/double word $< > 0$.

Hence, the correct answer is $ax = 003 CH$; $CF = 0$.

33. Which of the following regular expressions, each describing a language of binary numbers (MsB to LSB) that represents non-negative decimal values, does not include even values?

- (a) $0^*1+0^*1^*$ (b) 0^*10+1^*
(c) $0^*1^*0^*1+$ (d) $0^*1^*0^*1^*$

Where $\{+, *\}$ are quantification characters.

Ans : (c) When a number is represented in binary form, for add numbers, we have a 1 in the LSB while a 0 in LSB for even numbers

So, only the language generated by option (c) $0^*1^*0^*1+$ will, for sure, have 1 in the LSB, So this language won't contain even number.

34. Which of the following statements is/are TRUE?

(A) The grammar $S \rightarrow SS$ is ambiguous. (where S is the start symbol)/ τ

(B) The grammar $S \rightarrow 0S1 \mid 01S \mid \epsilon$ is ambiguous. (The special symbol ϵ represents the empty string) (Where S is the start symbol)/ τ

(C) The grammar (Where S is the start symbol)

$S \rightarrow T/U$

$T \rightarrow xSy \mid xy \mid \epsilon$

$U \rightarrow yT/\tau$

generates a language consisting of the string $yxxyy$.

- (a) Only (A) and (B) are TRUE
(b) Only (A) and (C) are TRUE
(c) Only (B) and (C) are TRUE
(d) All of (A), (B) and (C) are TRUE

Ans : (d) (a) $S \rightarrow SS \mid a$

This grammar is ambiguous. Take the string aaa for example to generate this string we can either have :

$S \rightarrow SS \rightarrow Sa \rightarrow SSa \rightarrow Saa \rightarrow aaa$

(b) $S \rightarrow 0S1 \mid 01S \mid \epsilon$

This grammar is also ambiguous. The string 01 have two different derivation trees.

$S \rightarrow 0S1 \rightarrow 01$ & $S \rightarrow 01S \rightarrow 01$

(c) This statement is TRUE. Here's how we can generate the string $yxxyy$ from the given grammar. $S \rightarrow U \rightarrow yT \rightarrow yxSy \rightarrow yxTy \rightarrow yxxyy$

So all the three statements are TRUE

35. Match the description of several parts of a classic optimizing compiler in List-I, with the names of those parts in List - II :

List-I		List-II	
A.	A part of a compiler that is responsible for recognizing syntax	i.	Optimizer
B.	A part of a compiler that takes as input a stream of characters and produces as output a stream of words along with their associated syntactic categories	ii.	Semantic Analysis
C.	A part of a compiler that understand the meanings of variable names and other symbols and checks that they are used in ways consistent with their definitions.	iii.	Parser
D.	An IR-to-IR transformer that tries to improve the IR program in some way (Intermediate Representation)	iv.	Scanner

Code :

	A	B	C	D
(a)	iii	iv	ii	i
(b)	iv	iii	ii	i
(c)	ii	iv	i	iii
(d)	ii	iv	iii	i

Ans : (a)

(i) **Parser** which is also called **syntax analyzer** who is responsible for recognizing syntax.

(ii) **Scanner** which is also called **lexical analyzer** who is responsible for tokenizing input stream of characters and produce tokens.

(iii) **Semantic analyzer** is part of compiler that understand the meaning of variable and check that they are used in same way consistent with their definitions.

(iv) **Optimizer** is responsible for **IR-IR transformation** that tries to improve the IR program in same way.

36. In distributed system, the capacity of a system to adapt the increased service load is called.....

(a) Tolerance

(b) Scalability

(c) Capability

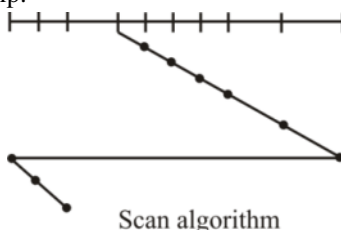
(d) Loading

Ans : (b) Scalability is an attribute that describes the ability of a process, network, software or organization to grow and manage increased demand a system, business that is described as scalable has an advantage because it is more adaptable to the changing needs & even presence of its users.

37. In disk scheduling algorithm, the disk head moves from one end to other end of the disk, serving the request along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without serving any requests on the return trip.

- (a) LOOK (b) SCAN
(c) C – LOOK (d) c – SCAN

Ans : (b) In scan disk scheduling algorithm, the disk head moves from one end to another end of disk, serving the requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk, without serving any request on the return trip.



38. Suppose there are six files F1, F2, F3, F4, F5, F6 with corresponding sizes 150 KB, 225 KB, 75 KB, 60 KB, 275 KB and 65 KB respectively. The files are to be stored on a sequential device in such a way that optimizes access time. In what order should the files be stored?

- (a) F5, F2, F1, F3, F6, F4
(b) F4, F6, F3, F1, F2, F5
(c) F1, F2, F3, F4, F5, F6
(d) F6, F5, F4, F3, F2, F1

Ans : (b) Files with sizes:

F1 = 150 KB, F2 = 225 KB, F3 = 75 KB,
F4 = 60 KB, F5 = 275 KB, F6 = 65 KB

So, to optimize access time merge two smallest files and put back in same bag. Repeatedly i.e.

(F4, F6) = 125 KB

Then ((F4, F6), F3) = 200 KB

Then, (((F4, F6), F3), F1) = 350 KB

Then, (F2, F5) = 500 KB

Then finally ((F4, F6, F3, F1), (F2, F5)) = 700 KB

So, order will be F4, F6, F3, F1, F2, F5.

39. Which module gives control of the CPU to the process selected by the sort - term scheduler?

- (a) Dispatcher (b) Interrupt
(c) Scheduler (d) Threading

Ans : (a) Dispatcher is the module that gives control of the CPU to the process selected by the sort term scheduler.

- 40. Two atomic operations permissible on Semaphores are and**
- (a) wait, stop (b) wait, hold
 - (c) hold, signal (d) wait, signal

Ans : (d) Wait & signal operations are the two atomic operations performed on a semaphore.

- 41. Software not wear-out in the traditional sense of the term, but software does tend to deteriorate as it evolves, because :**
- (a) Software suffers from exposure to hostile environments
 - (b) Defects are more likely to arise after software has been used often
 - (c) Multiple change requests introduce errors in component interactions
 - (d) Software spare parts become harder to order

Ans : (c) As time passes software is likely to get multiple change request while using software by users & one does not identify all aspects of software when it is developed.

- 42. Software re-engineering is concerned with :**
- (a) Re-constructing the original source code from the existing machine (low - level) code program and modifying it to make it more user - friendly
 - (b) Scrapping the source code of a software and re-writing it entirely from scratch
 - (c) Re-organizing and modifying existing software systems to make them more maintainable
 - (d) Translating source code of an existing software to a new machine (low - level) language

Ans : (c) Software re-engineering is a process of software development which is done to improve the maintainability of a software system.

- 43. Which of the following is not a key issue stressed by an agile philosophy of software engineering?**
- (a) The importance of self-organizing teams as well as communication and collaboration between team members and customers
 - (b) Recognition that change represents opportunity
 - (c) Emphasis on rapid delivery of software that satisfies the customer
 - (d) Having a separate testing phase after a build phase

Ans : (d) Principles of agile software development:

- 1. Customer satisfaction by early and continuous delivery of valuable software.
- 2. Welcome changing requirements, even in late development.
- 3. Working software is delivered frequently.

4. Close, daily cooperation between business people and developers.
 5. Face to face conversation is the best form of communication.
 6. Projects are build around motivated individuals who should we trusted.
 7. Working software is primary measure of progress.
 8. Sustainable development, able to maintain a constant pace.
 9. Continuous attention to technical excellence and good design.
 10. Simplicity the art of maximizing the amount of work not done is essential.
- So, (d) is best possible answer.

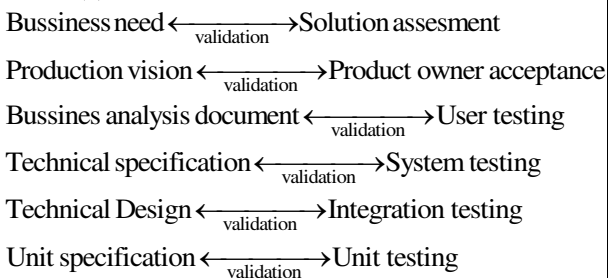
44. What is the normal order of activities in which traditional software testing is organized ?

- (A) Integration Testing
- (B) System testing
- (C) Unit Testing
- (D) Validation Testing

Code :

- (a) (C), (A), (B), (D)
- (b) (C), (A), (D), (B)
- (c) (D), (C), (B), (A)
- (d) (B), (D), (A), (C)

Ans : (a)



45. Which of the following testing techniques ensures that the software product runs correctly after the changes during maintenance?

- (a) Path Testing
- (b) Integration Testing
- (c) Unit Testing
- (d) Regression Testing

Ans : (d) Regression testing is a type of software testing which verifies that software which was previously developed and tested still performs the same way it was changed with other software. Changes may include software enhancements, patches etc.

46. Which of the following Super Computers in the fastest Super Computer?

- (a) Sun-way Taihu Light
- (b) Titan
- (c) Piz Daint
- (d) Sequoia

Ans : (a) Sunway Taihu Light has been named as the world's fastest supercomputer title, as per recently released semiannual top 500 list of super computers. It is developed by chain's National research centre of parallel computer engineering & technology (NRCPL).

47. Which of the following statements about ERP system is true?

- (a) Most ERP software implementations fully achieve seamless integration.
- (b) ERP software package are themselves combinations of separate applications for manufacturing, materials, resource planning, general ledger, human resources, procurement and order entry
- (c) Integration of ERP systems can be achieved in only one way
- (d) An ERP package implemented uniformly throughout an enterprise is likely to contain very flexible connections to allow charges and software variations.

Ans : (b) Some ERP software packages are themselves combinations for manufacturing, material resource planning, general ledger, human resources procurement & order entry.

48. Which of the following is not a clustering method?

- (a) K–Mean method
- (b) Self Organizing feature map method
- (c) K – nearest neighbor method
- (d) Agglomerative method

Ans : (c) In pattern recognition, the k-nearest neighbors algorithm (K-NN) is a non-parametric method used for classification and regression.

49. Which of the given wireless technologies used in lot, consumes the least amount of power?

- (a) Zigbee
- (b) Bluetooth
- (c) Wi-Fi
- (d) GSM/CDMA

Ans : (b) Bluetooth is a telecommunications industry specification that describes how mobile devices, computers and other device can easily communicate with each other using a short range wireless connection.

50. Which speed up could be achieved according to Amdahl's law of infinite number of processes if 5% of a program is sequential and the remaining part is ideally parallel?

- (a) Infinite
- (b) 5
- (c) 20
- (d) 50

Ans : (c) Sequential part – 5%
Parallel part (P) = 1 – Sequential part
= 1 – (5 ÷ 100)
= 1 – 0.05
= 0.95 or 95%
According to
Amdahl's Law,
$$S = \frac{1}{1 - 0.95} = 20$$

Hence, the value of speed up is 20.